Phase 2: Technical Application

EV Charging Energy Consumption Prediction

Machine Learning Coursework — Option A: Baseline Implementation

Research paper: Hussain et al. (2025), *Scientific Reports*, 15, Article 16124.

Evidence basis: submitted notebook and two CSV outputs. Numerical results describe the submitted run and must be refreshed after running the revised notebook.

Submitted-run results

Split / model	MAE (kWh)	RMSE (kWh)	R^2	Interpretation
Random — Dummy median	5.728	9.161	-0.0625	Lowest MAE on random split
Random — Random Forest	5.914	8.662	0.0501	Low positive R^2
Random — Extra Trees	5.897	8.649	0.0528	Lowest RMSE among main models
Chronological — Dummy median	6.422	10.200	-0.0636	Lowest chronological MAE
Chronological — Random Forest	6.741	9.908	-0.0035	Lowest chronological RMSE
Chronological — Extra Trees	6.751	9.933	-0.0086	Negative R^2
Chronological — proposed features	6.754	9.940	-0.0100	No improvement
Chronological — proposed + log1p	6.478	10.320	-0.0888	MAE improves, RMSE/R^2 worsen

1. Aim and scope

This Phase 2 implementation addresses Option A (Baseline Implementation) in the coursework brief. It uses Python and scikit-learn to build regression baselines for predicting EV charging-session energy consumption in kWh, then compares local results with the performance reported by Hussain et al. (2025).

The submitted notebook implemented a Dummy Median baseline, Random Forest, and Extra Trees using paper-style duration and calendar features. It also tested cyclical time features and a log1p transformation of the target. Two evaluation strategies were used: a random 80/20 split and a chronological 80/20 split.

2. Dataset and preprocessing

The experiment uses the City of Boulder EV charging-session dataset associated with the selected research paper. The notebook automatically detects the energy target, session start/end timestamps, total duration, and charging duration from column names, with manual overrides available when detection is incorrect. Invalid start timestamps, missing/non-numeric energy targets, and negative energy values are removed. Negative duration values are converted to missing values and median imputation is fitted inside model pipelines using training data.

The paper-style feature set contains total duration, charging duration, start hour, day of week, and month. The proposed feature set adds sine/cosine encodings for cyclical time variables and, when the required source columns are available, an idle ratio. The target is energy consumption in kWh. These choices broadly reflect the feature categories discussed in the research paper and Phase 1 review.

An important limitation is feature availability: total duration and charging duration may not be known when a prediction is required before a session begins. Therefore, this implementation should be described as reproducing the paper-style feature setup, not as proving reliable pre-session forecasting.

3. Models, evaluation and reproducibility

The baseline is DummyRegressor with the median strategy. The two machine-learning models are RandomForestRegressor and ExtraTreesRegressor. The tree models use 200 estimators and fixed random_state=42; the Random Forest uses min_samples_leaf=2 and Extra Trees uses min_samples_leaf=1. Missing feature values are imputed with training-set medians through scikit-learn pipelines.

MAE measures average absolute prediction error; MSE and RMSE penalise larger errors, with RMSE expressed in kWh; R^2 compares model predictions against the test-target mean reference; and NRMSE is defined in the notebook as RMSE divided by the range of target values in the test set. The NRMSE definition must be stated because it may not be directly

comparable with a paper if the normalisation denominator differs.

The random split provides a comparison approximating the paper's random hold-out setup. The chronological split tests later records and is more informative for future-period generalisation. However, the current CSVs were generated by the earlier notebook version, which split sorted rows rather than keeping all identical timestamps together. The revised notebook fixes this and adds timestamp-grouped temporal cross-validation; it must be rerun before the final numbers are treated as the revised experiment's final results.

4. Results from the submitted run

The submitted result CSV contains nine experiment rows. The following table summarises the three main models on the two original splits. These values are copied from the submitted CSV, not newly computed.

5. Interpretation

On the random 80/20 split, Extra Trees had RMSE 8.649 kWh and R^2 0.0528, while Random Forest had RMSE 8.662 kWh and R^2 0.0501. Extra Trees therefore had the lowest RMSE and highest R^2 among the two learned models, but the Dummy Median baseline achieved a lower MAE (5.728 kWh) than either learned model. R^2 was close to zero, indicating weak explanatory performance.

On the original chronological split, Random Forest had the lowest RMSE among the main chronological models (9.908 kWh), while the Dummy Median had the lowest MAE (6.422 kWh). All chronological R^2 values were negative. This means the models did not outperform the test-set mean reference under R^2. The results suggest that generalisation to later records is difficult, but they do not establish the cause. Target distribution shift, feature limitations, duplicate records, or other data properties should be investigated.

The proposed Extra Trees feature set did not improve the reported chronological metrics over the paper-style feature set. The log1p-target variant reduced MAE from 6.751 to 6.478 kWh, but increased RMSE from 9.933 to 10.320 kWh and lowered R^2 from -0.0086 to -0.0888. Therefore, the transformation did not produce an overall improvement across the reported metrics. The model with the best metric depends on which metric is prioritised; a single metric should not be used to claim universal superiority.

6. Comparison with Hussain et al. (2025)

The research article reports Extra Trees results of MAE 0.5888 kWh, MSE 3.2683 kWh^2, R^2 0.9592, RMSE 1.8078 kWh, and NRMSE 0.0200 in its abstract. These results are substantially stronger than the submitted local results. This difference must be discussed cautiously: the current implementation uses a simplified feature set, fixed baseline parameters, and a chronological evaluation in addition to the random split. Direct comparison requires matching the exact dataset version, cleaned records, feature definitions, target, preprocessing, hyperparameters, split strategy, and metric definitions.

The Phase 1 report's comparison table lists Extra Trees MAE as 0.82 kWh, while the paper abstract and the notebook's reference table report 0.5888 kWh. This is an internal reporting inconsistency that should be corrected in the final coursework: use the paper's original value of 0.5888 kWh when citing the abstract, unless another specific table in the article is being cited and explained.

7. Data-quality and implementation issues found

The submitted chronological prediction CSV contains 28,064 rows. In the exported columns (timestamp, actual target and model predictions), 6,372 rows are exact duplicates and 7,398 timestamps repeat. This does not prove that the raw dataset has duplicate sessions, because the export does not include station/session identifiers or every original column; separate source records can share the exported values. The revised notebook therefore audits exact duplicates across all original raw columns, preserves a source-row ID, and does not automatically delete duplicates without investigation.

The earlier notebook selected the diagnostic model by the lowest RMSE on the chronological test set. Choosing a model based on final test performance and then presenting that same set as an independent diagnostic can bias interpretation. The revised notebook fixes the diagnostic model in advance to the paper-style Extra Trees candidate. It also keeps identical timestamps together across the chronological split and temporal-CV folds. These changes may alter metrics, so the revised notebook should be run from top to bottom before submission.

8. Conclusion and next steps

The submitted implementation demonstrates a reproducible baseline workflow and a comparison between random and chronological evaluation. However, the reported R^2 values are weak, and the learned models do not consistently outperform the Dummy baseline on every metric. The experiments therefore do not yet support a claim of strong predictive performance.

Before final submission, run the revised notebook against the original dataset, confirm the detected column mapping and duplicate audit, inspect the chronological train/test periods, and replace the old CSVs with the newly generated outputs. Preserve the new results without manually editing scores. Report the actual outcomes, limitations, and the distinction between paper-reported and locally measured performance.

References

Hussain, I., Ching, K. B., Uttraphan, C., Tay, K. G., & Noor, A. (2025). Evaluating machine learning algorithms for energy consumption prediction in electric vehicles: A comparative study. *Scientific Reports, 15*, Article 16124. https://doi.org/10.1038/s41598-025-94946-7